

MATH 2130 - Week 3

Inner Products and Orthogonality

1 Overview

In order to further understand the idea of orthogonality, we will begin this week with the idea of an **inner product** on a vector space. We will use these to define the **norm** of a vector and define the **angle** between two elements of a vector space. We will also use inner products to define the **projection** of a vector onto another and the projection of a vector onto a subspace.

Once we have these tools, we will combine these ideas with the ideas from week 2 and discuss **orthogonal bases** and **orthogonal matrices**. We will learn an algorithmic way to turn any basis into an **orthonormal basis**. This algorithm is called **Gram-Schmidt Orthogonalization**.

After this, we will talk briefly about the **determinant** of an $n \times n$ matrix. We will give explicit formulas for determinants of “small” matrices and talk about ways to find determinants of square matrices of any size. We will list some of the basic properties of matrices and their applications.

We will have a midterm exam on Friday! This will include all material covered in the course except determinants.

2 Schedule and Checklist

Here is a brief schedule for the week. Please note that this is subject to change, so you need to adjust to what we do in class, regardless of what it says here. Think of this as a pretty accurate guideline, though.

- (M) Inner Products and Projections.
- (T) Norms and Angles. Projection onto Subspaces.
- (W) Orthogonal Bases and Gram-Schmidt Orthogonalization.
- (R) Determinants. Review for Midterm.
- (F) **Midterm Exam**

Here is a checklist of things to do this week. This is meant to help you keep track of what you’ve completed and what you still need to work on.

- ☐ Read these notes and ask any questions that you have about anything.
- ☐ Read the sections from Austin and Olver.
- ☐ Complete all of the homework exercises.

3 Key Concepts

- Inner Products
- Norm and Angle
- Unit Vectors and Normalization
- Projection onto a vector and a subspace
- Orthogonal Bases and Orthogonal Matrices
- Gram-Schmidt Orthogonalization
- Determinants

4 Important Definitions, Formulas, and Theorems

- Consider two vectors in the plane, $a = (a_1, a_2)$ and $b = (b_1, b_2)$. Let α be the angle that the vector a makes with the positive x -axis, and let β be the angle that the vector b makes with the positive x -axis. Let $\theta = \beta - \alpha$ be the angle between the two vectors (assume, without loss of generality, that $\beta > \alpha$. (You really should draw a picture of this. For simplicity, let both a and b be in the first quadrant.)

By basic trigonometry, we have the following:

$$\sin \alpha = \frac{a_2}{\|a\|}, \quad \text{and} \quad \cos \alpha = \frac{a_1}{\|a\|}$$

$$\sin \beta = \frac{b_2}{\|b\|}, \quad \text{and} \quad \cos \beta = \frac{b_1}{\|b\|}$$

Using the addition/subtraction identity for cosine, we have

$$\cos \theta = \cos(\beta - \alpha) = \cos \beta \cos \alpha + \sin \beta \sin \alpha = \frac{a_1 b_1 + a_2 b_2}{\|a\| \|b\|}$$

We define the numerator of the right-hand side, $a_1 b_1 + a_2 b_2$ to be the **inner product** of the two vectors a and b :

$$a \cdot b = (a_1, a_2) \cdot (b_1, b_2) = a_1 b_1 + a_2 b_2$$

Alternate notation for the inner product is $\langle a, b \rangle$. Please note that if a and b are written as column vectors, as is customary, then

$$a \cdot b = [a_1, a_2] \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = a^T b$$

Finally, note that the **norm** of the vector a (or any 2-dimensional vector, in general) can be given in terms of the inner product of the vector a with itself:

$$\|a\| = \sqrt{a_1^2 + a_2^2} = \sqrt{a \cdot a} = \sqrt{\langle a, a \rangle}$$

- Everything in the preceding item (except the intuitive geometric and trigonometric visualization) generalizes immediately to the case of n -dimensional vectors.

If $a = (a_1, a_2, \dots, a_n)$ and $b = (b_1, b_2, \dots, b_n)$, then

$$a \cdot b = \langle a, b \rangle = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = \sum_{i=1}^n a_i b_i = a^T b$$

and

$$\|a\| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2} = \sqrt{a \cdot a} = \sqrt{\langle a, a \rangle}$$

Finally, we define the cosine of the angle, call it θ , between any two n -dimensional vectors a and b to be

$$\cos \theta = \frac{a \cdot b}{\|a\| \|b\|}$$

- Two vectors a and b are **orthogonal** if and only if $a \cdot b = 0$.
- The **projection** of the vector b onto (the line spanned by) the vector a is given by

$$\text{proj}_a b = \frac{a \cdot b}{a \cdot a} a = \frac{a \cdot b}{\|a\|^2} a$$

- Any two vectors satisfy the **Cauchy-Schwarz Inequality**:

$$|a \cdot b| \leq \|a\| \|b\|$$

Equality holds (i.e. $|a \cdot b| = \|a\| \|b\|$) only when $b = ka$ for some constant k .

- Let x be an n -dimensional vector, and let y be an m -dimensional vector, and let A be a $m \times n$ matrix. Then the **transpose** of A , denoted by A^T , is defined by the following property:

$$\langle Ax, y \rangle = \langle x, A^T y \rangle$$

Practically speaking, the rows of A^T are the columns of A , and the columns of A^T are the rows of A . So the transpose of a matrix interchanges rows and columns. In terms of the entries, we have $(A^T)_{ij} = (A)_{ji}$.

- If A and B are compatible matrices (so the product AB is defined), then $(AB)^T = B^T A^T$.
- A square matrix A is **symmetric** if $A^T = A$.
- If A is a $m \times n$ matrix of rank r , then $A^T A$ is a symmetric $n \times n$ matrix of rank r .
- If the $m \times n$ matrix A has linearly independent columns (so that the rank of A is $r = n$), then $A^T A$ is a square, symmetric, invertible $n \times n$ matrix.

- Let A be an $m \times n$ matrix and let b be any n -dimensional column vector. The linear system given by $Ax = b$ is **inconsistent** if the vector b does **not** lie in the column space of A (this means that b can not be written as a linear combination of the columns of A). In this case, the **normal equations** for the **least squares solution** to $Ax = b$ is given by

$$A^T A \bar{x} = A^T b$$

If the columns of A are linearly independent, then by the previous item the matrix $A^T A$ is invertible and we find that

$$\bar{x} = (A^T A)^{-1} A^T b$$

and the **projection** of b onto the column space of A is given by

$$p = A\bar{x} = A(A^T A)^{-1} A^T b$$

Geometrically, p is the point in the column space of A that is “closest” to the vector b . The vector starting at p and ending at b is orthogonal to the column space of A .

- The **projection matrix** $P = A(A^T A)^{-1} A^T$ has two basic properties:

1. It is **idempotent**: $P^2 = P$
2. It is symmetric: $P^T = P$

Conversely, any matrix with these two properties is a projection matrix onto the column space of P .

- A set of vectors $v_1, v_2, \dots, v_k$ is called **orthonormal** if

$$v_i \cdot v_j = \begin{cases} 0 & \text{if } i \neq j \text{ (giving the orthogonality)} \\ 1 & \text{if } i = j \text{ (giving the normalization)} \end{cases}$$

- If the basis $B = \{b_1, b_2, \dots, b_n\}$ is an orthonormal set of vectors, then B is called an **orthonormal basis**.
- Let A be an $m \times n$ matrix, and suppose that the columns of A form an orthonormal set of vectors. Then the columns are linearly independent and we have

$$A^T A = I_n$$

where I_n is the $n \times n$ identity matrix.

In this case the projection matrix onto the column space of A becomes $P = AA^T$ and the projection of b onto the column space of A becomes $p = AA^T b$.

- Let Q be a square matrix with orthonormal columns. Then Q is called an **orthogonal matrix** (note that the name is *orthogonal* even though the columns are *orthonormal*).
- Let Q be an $n \times n$ orthogonal matrix. Then Q has the following properties:
 1. $Q^T Q = I_n$

2. $QQ^T = I_n$
 3. $Q^T = Q^{-1}$
 4. $\|Qx\| = \|x\|$ for any vector x (Q “preserves lengths”).
 5. $\langle Qx, Qy \rangle = \langle x, y \rangle$ for any two vectors x, y (Q “preserves inner products”).
- Any set of linearly independent vectors $a_1, a_2, \dots, a_n$ can be converted into a set of orthonormal vectors by the **Gram-Schmidt Orthogonalization** algorithm (also called the Gram-Schmidt Process), which proceeds in two steps.

First, compute the set of orthogonal vectors $v_1, v_2, \dots, v_n$ in the following way:

$$v_1 = a_1$$

$$v_i = a_i - \sum_{k=1}^{i-1} \frac{\langle v_k, a_i \rangle}{\langle v_k, v_k \rangle} v_k$$

The vectors $v_1, v_2, \dots, v_n$ are an orthogonal set of vectors that have the same span as the original set of vectors $a_1, a_2, \dots, a_n$.

Second, normalize the vectors v_i :

$$q_i = \frac{v_i}{\|v_i\|}$$

The set of vectors $q_1, q_2, \dots, q_n$ is an orthonormal set of vectors.

The most common case is that $B = \{a_1, a_2, \dots, a_n\}$ is a basis for the vector space V . Then the Gram-Schmidt Process applied to this basis produces $B' = \{q_1, q_2, \dots, q_n\}$, which is an orthonormal basis for V .

You will **NEVER** want to do this by hand (except perhaps in the 2-dimensional case). Sage is your friend here.

- Any matrix A with linearly independent columns can be factored into $A = QR$. The columns of Q are orthonormal, and R is upper triangular and invertible. If the original matrix A is square, then so are its two factors Q and R , and Q becomes an orthogonal matrix. This is called the **QR Decomposition** or the **QR Factorization** of the matrix A .

4.1 Determinants

If A is an $n \times n$ matrix, then the **determinant** of A , $\det(A)$ is a function that maps A to a *very* complicated but important combination of the entries of A .

Before describing how to find determinants (for small matrices), we have the following:

- **The Most Important Fact** If $\det(A) \neq 0$ the A is *nonsingular*. There are dozens of statements that are equivalent to this. Probably the most important is that if A is nonsingular then it is invertible.
- **The Most Important Application** We will use determinants to find eigenvalues.
- **Second Most Important Fact** Given two $n \times n$ matrices A and B , then $\det(AB) = \det(A) \det(B)$.

- **Geometric Interpretation** If the columns of A are treated as n -dimensional vectors, then these vectors form a parallelepiped (an n -dimensional parallelogram) in n -dimensional space. The hyper-volume (higher dimensional equivalent of area/volume) of this object is given by $|\det(A)|$. The absolute values are there because determinants can be negative and hyper-volume is a nonnegative value. The geometric interpretation of the negative sign is that certain transformations “reverse the orientation” of the starting shape (think of picking a parallelogram up out of the plane, flipping it over and setting it back down).

- **Determinants of Special Matrices**

- $\det(I_n) = 1$.
- Let E_1 be an elementary matrix that adds multiple of one row to another, then $\det(E_1) = 1$.
- Let E_2 be an elementary matrix that swaps one row with another, then $\det(E_2) = -1$.
- Let E_3 be an elementary matrix that scales one row by c , then $\det(E_3) = c$.
- If D is a diagonal matrix with entries d_i , then $\det(D) = d_1 d_2 \cdots d_n$.
- If U is an upper triangular matrix, then $\det(U) = a_{11} a_{22} \cdots a_{nn}$.
- If L is a lower triangular matrix, then $\det(L) = a_{11} a_{22} \cdots a_{nn}$.
- If $\det(A) \neq 0$ then $\det(A^{-1}) = 1/\det(A)$.
- If $\det(S) \neq 0$ then $\det(S^{-1}AS) = \det(A)$.
- $\det(A^T) = \det(A)$

Okay, how do we calculate determinants? The short answer is you use a computer. The only time you will ever compute a determinant by hand is when A is 2×2 (okay, sometimes you will calculate 3×3 determinants by hand). When A is 2×2 , the formula is pretty simple and you should definitely memorize this.

$$\text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ then } \det(A) = ad - bc$$

If you have to know, here’s the formula for a general $n \times n$ matrix. Let A_{ij} be the $(n-1) \times (n-1)$ matrix you get by crossing out the i -th row and j -th column from A . This is called a minor of A . Then define the cofactor c_{ij} by the following formula:

$$c_{ij} = (-1)^{i+j} \det(A_{ij})$$

Once you’ve done this, then

$$\det(A) = \sum_{j=1}^n a_{ij} c_{ij}$$

for any row i of A . This is a recursive definition called either the Cofactor Expansion, the Laplace Expansion, or Expansion by Minors. You will **NEVER** actually use this as a computational tool. In practice, you will use Gaussian Elimination (surprise, surprise).

5 Examples

- Consider the following 3×2 matrix

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 0 \\ 3 & 1 \end{bmatrix}$$

The columns are linearly independent (this is easy to “see”) so the rank of A is 2. We then have

$$A^T A = \begin{bmatrix} 14 & 7 \\ 7 & 17 \end{bmatrix}$$

which is a 2×2 symmetric matrix with $\det A^T A = (14)(17) - (7)(7) \neq 0$, so it is also invertible.

- Let the matrix A and vector b be given by

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 5 \\ 0 & 0 \end{bmatrix}, \quad \text{and} \quad b = \begin{bmatrix} 4 \\ 3 \\ 9 \end{bmatrix}$$

Then we have

$$A^T A = \begin{bmatrix} 2 & 7 \\ 7 & 29 \end{bmatrix}, \quad (A^T A)^{-1} = \frac{1}{9} \begin{bmatrix} 29 & -7 \\ -7 & 2 \end{bmatrix}$$

and we find that

$$p = A(A^T A)^{-1} A^T b = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$$

- If b is in the column space of A then there exists a vector x such that $b = Ax$ (the right-hand side is giving a linear combination of the columns of A). Then we have that

$$p = A(A^T A)^{-1} A^T b = A(A^T A)^{-1} A^T Ax = Ax = b$$

so the projection is just the original vector b . What this is saying geometrically is that the closest point to b is b itself, which is obvious (but only true when b is in the column space of A).

- The simplest example of an orthogonal matrix is I_n , the identity matrix.
- The simplest non-trivial 2×2 example of an orthogonal matrix is

$$Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Geometrically, this matrix rotates any vector by an angle θ counter-clockwise around the origin. It is an example of a **rotation matrix**.

- Let P be any $n \times n$ matrix that has only a single 1 in every row and every column. This matrix is called a **permutation matrix** because it can be formed from the identity matrix by permuting the columns (or the rows). Every permutation matrix is orthogonal.

- Suppose you have the following three vectors:

$$a_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad a_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

The easy one is $v_1 = a_1$. For v_2 we have

$$\frac{\langle a_2, v_1 \rangle}{\langle v_1, v_1 \rangle} = \frac{1}{2}, \quad \text{which gives } v_2 = a_2 - \frac{1}{2}v_1 = \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}$$

For v_3 we have

$$\frac{\langle a_3, v_1 \rangle}{\langle v_1, v_1 \rangle} = \frac{1}{2}, \quad \frac{\langle a_3, v_2 \rangle}{\langle v_2, v_2 \rangle} = \frac{1}{3}, \quad \text{which gives } v_3 = a_3 - \frac{1}{2}v_1 - \frac{1}{3}v_2 = \begin{bmatrix} -2/3 \\ 2/3 \\ 2/3 \end{bmatrix}$$

These vectors are orthogonal, but they are not normalized. So we do that now:

$$q_1 = \frac{v_1}{\|v_1\|} = \sqrt{\frac{1}{2}}v_1, \quad q_2 = \frac{v_2}{\|v_2\|} = \sqrt{\frac{2}{3}}v_2, \quad \text{and} \quad q_3 = \frac{v_3}{\|v_3\|} = \sqrt{\frac{3}{4}}v_3$$

You can check that these are an orthonormal basis for $\mathbb{R}^3$.

If we work backwards, we can find the QR Factorization for the matrix A that has the a_i as columns. First, start by writing the a 's as linear combinations of the q 's (which you must be able to do since $\{q_1, q_2, q_3\}$ is a basis). This gives

$$a_1 = \sqrt{2}q_1$$

$$a_2 = \sqrt{\frac{1}{2}}q_1 + \sqrt{\frac{3}{2}}q_2$$

$$a_3 = \sqrt{\frac{1}{2}}q_1 + \sqrt{\frac{1}{6}}q_2 + \sqrt{\frac{4}{3}}q_3$$

This can be written in matrix notation as follows:

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} = \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix} \begin{bmatrix} \sqrt{2} & \sqrt{\frac{1}{2}} & \sqrt{\frac{1}{2}} \\ 0 & \sqrt{\frac{3}{2}} & \sqrt{\frac{1}{6}} \\ 0 & 0 & \sqrt{\frac{4}{3}} \end{bmatrix}$$

The first matrix on the right (with columns given by q_i) is our orthogonal matrix Q , and the second matrix on the right (with the weights from the linear combinations) is our upper triangular matrix R .

6 Sage Code

6.1 Commands

These are the most common Sage commands that we will use this week. Think of this more as a reference. It is usually easier to understand what the commands do by seeing them used in example code, which can be found in the Google Drive.

- `.dot_product()`
- `.inner_product()`
- `.norm()`
- `.normalized()`
- `.gram_schmidt()`

6.2 Example Code

See the documents in the Google Drive folder for example code.

7 Additional Comments

- It's hard to overemphasize how important the inner product is. It gives vector spaces (and everything connected to them) a geometric structure. This is huge.
- There are infinitely many different inner products on any vector space. What we've been calling the inner product is more properly called the standard inner product. We have defined it as $v^T w$ which is really $v^T I w$, with the $n \times n$ identity matrix. This can be generalized to any product $v^T A w$, provided the matrix A has certain 'nice' properties.
- It is also hard to overemphasize the importance of the concept of orthogonality. This shows up everywhere in more advanced mathematics. The main advantage of working with objects that are orthogonal is that they simplify computations immensely, since lots of things become 0. Anything that simplifies computation, and clears away all that is inessential from what is essential, is worth its weight in gold.

8 Reading Notes

Please note: there are two pdfs for this week. The first contains the material connected to Inner Products and Orthogonality. The second, which is much shorter (only a few pages) contains material related to determinants.

- Olver 3.1 - Inner Products
- Olver 3.2 - Inequalities
- Olver 3.3 - Norms
- Olver 4.1 - Orthogonal and Orthonormal Bases
- Olver 4.2 - The Gram-Schmidt Process
- Olver 4.3 - Orthogonal Matrices
- Olver 4.4 - Orthogonal Projection and Orthogonal Subspaces
- Olver 1.9 - Determinants

9 Homework Exercises

Please see the handout for the Week 3 homework problems (there will also be a file in the Google Drive for this).